

Independent Replication Report: Altermagnetic spin textures coupled to superconductors — Domain-wall spin-triplet superconductivity

REPLICATE-PROJECT / TEXTURES-100
 Paper: Dar, Scheurer, Schrade, arXiv:2607.15249v1

July 20, 2026

Abstract

We independently rebuild, from scratch (no author code), the real-space Bogoliubov–de Gennes (BdG) calculation behind the central pairing-correlation result of Dar, Scheurer & Schrade, “Altermagnetic spin textures coupled to superconductors” (arXiv:2607.15249v1). A conventional s -wave superconductor is proximitized by a circular altermagnetic Néel domain wall; the projected low-energy Hamiltonian carries an emergent radial spin–orbit field $\alpha(\mathbf{r})$ and an altermagnet-only emergent Zeeman field $V_z(\mathbf{r}, \chi)$. We diagonalize the $4N^2 = 7744$ -dimensional BdG matrix on a 44×44 grid (~ 69 nm box) and extract the on-site singlet amplitude and the equal-spin p -wave triplet bond intensity $I_t(\mathbf{r})$.

VERDICT: PARTIAL (Coverage $\approx 5/10$, Agreement $\approx 6/10$)

The mechanism reproduces: triplets are generated *only* at the wall (localization fraction 0.99) and the spin-resolved intensities have the correct angular character ($I_t^{\uparrow\uparrow}$ peaks near $\chi = \pm\pi/2$, $I_t^{\downarrow\downarrow}$ near $\chi = 0, \pi$). The *discriminating* headline claim — a fourfold angular hotspot modulation *unique to the altermagnet* that *vanishes* in the antiferromagnetic (AFM) limit $\varrho_z = 0$ — is *not* reproduced: the on-wall angular modulation is 0.119 (AM) vs 0.122 (AFM), essentially identical, and the fourfold ($\ell = 4$) Fourier power is 2.88×10^{-11} (AM) vs 3.00×10^{-11} (AFM). The triplet signal is also only 0.38% of the singlet, so this is a difference-of-small-numbers regime.

1 The paper’s claims (the comparison anchors)

1. **Triplet generation is localized at the wall.** Equal-spin triplets arise only where the emergent SOC $\alpha_p(\mathbf{R}) = \hbar\varrho_3\phi'(R)p_R$ is nonzero, i.e. where $\phi'(R) \neq 0$ (the wall). [Sec. VI, Eq. 15; Fig. 4a,b]
2. **Fourfold hotspot modulation, unique to the altermagnet.** In the AM case ($\varrho_z \neq 0$) the on-wall triplet intensity $I_t(\chi)$ forms a fourfold “hotspot” pattern; *this angular modulation disappears in the AFM limit $\varrho_z = 0$* (Fig. 4a vs 4b). This is the sharpest, most falsifiable claim.
3. **Spin-resolved anisotropy.** $I_t^{\uparrow\uparrow}$ peaks near $\chi = \pm\pi/2$; $I_t^{\downarrow\downarrow}$ peaks near $\chi = 0, \pi$; the total is enhanced where they overlap (Fig. 4c,d).
4. **d -wave form factor.** The AM node condition is a spin-dependent ellipse set by the d -wave $V_z \propto \cos 2\chi$; the AFM limit gives a circle.

5. **(Scoped out)** Supercurrent-induced quadrupolar torque (Sec. VII); nodal $\leftrightarrow$ fully-gapped transitions; the winding of the admixed triplet's momentum dependence.

2 Governing equations reimplemented

We implement the projected low-energy effective Hamiltonian for a planar radial Néel domain wall (paper Eq. 3/8), with $\hbar = 1$, energies in eV, lengths in nm:

$$\hat{h}_{\text{proj}}(\mathbf{r}, \mathbf{p}) = \xi(\mathbf{r}, \mathbf{p}) \sigma_0 + b_z(\mathbf{r}, \mathbf{p}) \sigma_z + \frac{1}{2} \{p, \alpha(\mathbf{r})\} \sigma_x, \quad (1)$$

$$\xi = \varrho_0 p^2 + \varrho_0 V_0(\mathbf{r}) - \mu, \quad V_0 = \frac{1}{4} \phi'(r)^2, \quad (2)$$

$$b_z = \varrho_z (p_x^2 - p_y^2) + \varrho_z V_z, \quad V_z = \frac{1}{4} \phi'(r)^2 \cos 2\chi, \quad (3)$$

$$\alpha(\mathbf{r}) = \varrho_3 \phi'(r) \hat{\mathbf{r}}, \quad \phi(r) = \frac{\pi}{2} \tanh((r - R_0)/w). \quad (4)$$

A fixed uniform proximity singlet Δ_0 (semiclassical, non-self-consistent) closes the BdG block (paper Eq. 7):

$$H_{\text{BdG}} = \begin{pmatrix} \hat{h}_{\text{proj}} & \Delta_0(-i\sigma_y) \\ -\Delta_0(-i\sigma_y) & -\hat{h}_{\text{proj}}^* \end{pmatrix}. \quad (5)$$

Kinetic terms use second-order central finite differences on the grid; the emergent SOC operator $\frac{1}{2} \{p, \alpha\}$ is symmetrized to enforce hermiticity. From the anomalous block $F_{ab} = \langle c_a c_b \rangle = \sum_{E>0} u_a v_b^*$ we extract the on-site singlet $\psi_s(\mathbf{r}) = |\frac{1}{2}(\langle c_\uparrow c_\downarrow \rangle - \langle c_\downarrow c_\uparrow \rangle)|$ and the equal-spin p -wave bond intensity $I_t(\mathbf{r}) = \sum_\sigma \sum_{\delta \in \{\hat{x}, \hat{y}\}} |\langle c_{\mathbf{r}\sigma} c_{\mathbf{r}+\delta, \sigma} \rangle|^2$ (a discrete realization of paper Eq. 16).

3 Methods / parameters

Symbol	Value	Symbol	Value
ϱ_0	1.0 eV nm ²	μ	0.01 eV
ϱ_z (AM)	0.1 eV nm ²	Δ_0	5×10^{-4} eV
ϱ_z (AFM)	0	R_0	20 nm
ϱ_3	0.05 eV nm ²	w	4 nm
grid N	44	spacing a	1.6 nm
box	68.8 nm	BdG dim	7744

Table 1: Parameters. Geometry is shrunk relative to the paper's $R \approx 50$ nm so the wall sits well inside a dense-diagonalizable box; the physics (wall-localized triplets, d -wave V_z) is scale-controlled by $\phi'(r)$ and is preserved.

4 Results

4.1 What reproduced

Claims 1 and 3 reproduce cleanly and qualitatively. Triplet intensity is overwhelmingly concentrated in the annulus around R_0 (localization fraction 0.99), exactly as Eq. (15) demands ($\alpha_p \propto \phi'(R)$ vanishes away from the wall). The spin-resolved intensities show the paper's Fig. 4(c) anisotropy: $I_t^{\uparrow\uparrow}$ is largest near $\chi = \pm\pi/2$ and $I_t^{\downarrow\downarrow}$ near $\chi = 0, \pi$ (see `angular_It_up_AM` / `angular_It_dn_AM` in the evidence JSON), consistent with the emergent-SOC triplet-generation mechanism.

Quantity (this work)	AM ($\varrho_z = 0.1$)	AFM ($\varrho_z = 0$)	Paper's claim / status
Hermiticity residual	0.0		clean (sanity)
Min $ E_{\text{BdG}} $ (gap)	5.1×10^{-4} eV		gapped, $\sim \Delta_0$ (OK)
Singlet max ψ_s	2.44×10^{-3}	—	finite s -wave (OK)
Triplet I_t max	8.52×10^{-11}	8.42×10^{-11}	finite triplet (OK)
$\sqrt{I_t^{\text{max}}}/\psi_s^{\text{max}}$	0.0038	—	triplet $\ll$ singlet
Triplet loc. frac ($ r - R_0 < 2w$)	0.992	0.993	localized at wall (MATCH, claim 1)
On-wall ang. modulation	0.119	0.122	should VANISH in AFM — FAILS (claim 2)
$\ell = 4$ (fourfold) FFT power	2.88×10^{-11}	3.00×10^{-11}	should be AM-only — FAILS (claim 2)
$\ell = 1$ FFT power	2.30×10^{-11}	2.23×10^{-11}	p -wave/dipolar background
$I_t^{\uparrow\uparrow}$ peak	near $\chi = \pm\pi/2$	—	MATCH (claim 3)
$I_t^{\downarrow\downarrow}$ peak	near $\chi = 0, \pi$	—	MATCH (claim 3)

Table 2: This-work vs paper. Bold rows are the direct headline comparisons.

4.2 What did NOT reproduce (the discriminating claim)

The paper's sharpest, most falsifiable statement — that the fourfold angular modulation is *unique to the altermagnet and disappears when $\varrho_z = 0$* — is not seen. Our AM and AFM on-wall $I_t(\chi)$ profiles are nearly identical: modulation depth 0.119 vs 0.122, and fourfold Fourier power 2.88×10^{-11} vs 3.00×10^{-11} . In our discretization the angular structure is dominated by the emergent SOC $\alpha \propto \hat{\mathbf{r}}$ (present in *both* AM and AFM), while the altermagnet-only $V_z \propto \cos 2\chi$ contributes a change of $\lesssim 3\%$ — below the level at which we can claim a clean AM-only fourfold pattern. Compounding this, $\sqrt{I_t}/\psi_s \approx 0.4\%$: the triplet is a tiny admixture on top of the singlet, so the AM/AFM *difference* is a difference of small numbers (cf. background-subtraction pitfall in the replication skill).

5 Comparison & verdict

PARTIAL. The proximity-induced, wall-localized equal-spin triplet mechanism and its spin-resolved anisotropy reproduce; the emergent SOC does generate triplets exactly where the paper says. But the *defining* altermagnet signature — a fourfold hotspot pattern that is absent in the AFM limit — is not resolved by this semiclassical, non-self-consistent, coarse-grid BdG implementation. We therefore cannot confirm the headline as stated, and we are honest that the observed AM/AFM difference lives in the noise of a difference-of-small-numbers.

6 Honest gaps (numbered; see `failure_analysis.md`)

- Fourfold-modulation null result.** AM $\approx$ AFM angular structure; V_z 's d -wave contribution is $\lesssim 3\%$, not the dominant, AFM-vanishing pattern the paper reports.
- Difference of small numbers.** Triplet is 0.4% of singlet; the AM–AFM contrast is not on statistically firm ground at this resolution.
- Coarse grid / small box.** $a = 1.6$ nm, $R_0 = 20$ nm (vs paper ~ 50 nm); the d -wave anisotropy on the wall may be under-resolved.
- Semiclassical, non-self-consistent Δ .** Uniform fixed Δ_0 ; the paper's Wigner-transform / self-consistent treatment (Eqs. 15–16) is approximated by a discrete bond correlator.

5. **Scoped out entirely:** supercurrent-induced quadrupolar torque (Sec. VII), nodal $\leftrightarrow$ gapped transitions, triplet momentum winding.
6. **Extraction tooling degraded (NOT physics):** `marker/nougat` absent; interim `pdftotext`. `pdflatex` absent; this `.tex` ships as source.

7 Reproducibility

Interpreter: `/home/stevens/comfyui-env/bin/python` (numpy 2.3.5, scipy 1.17.0). Solver: `report/evidence/dar2026_result.json`.
Evidence: `report/evidence/dar2026_result.json`.

```
cd work && /home/stevens/comfyui-env/bin/python dar2026_bdg.py
# ~125 s; writes dar2026_result.json (AM + AFM runs; 7744-dim dense eigh x2)
```

Live re-run during packaging matched the saved JSON to all quoted digits (hermiticity 0.0, singlet max 2.437×10^{-3} , I_t max AM/AFM $8.52/8.42 \times 10^{-11}$, modulation 0.119/0.122).